

Table of Contents

Counting Problems

1. 3413 — *Filled Subgrid Count I (no solution found)*
2. 3414 — *Filled Subgrid Count II (no solution found)*
3. 3415 — *All Letter Subgrid Count I (no solution found)*
4. 3416 — *All Letter Subgrid Count II (no solution found)*
5. 3417 — *Border Subgrid Count I (no solution found)*
6. 3418 — *Border Subgrid Count II (no solution found)*
7. 3400 — *Raab Game II (no solution found)*
8. 1080 — Empty String
9. 2229 — Permutation Inversions
0. 2176 — Counting Bishops
1. 2228 — Counting Sequences
2. 1078 — Grid Paths II
3. 1075 — Counting Permutations
4. 2429 — Grid Completion
5. 2421 — *Counting Reorders (no solution found)*
6. 3232 — *Tournament Graph Distribution (no solution found)*
7. 3157 — *Collecting Numbers Distribution (no solution found)*
8. 2415 — Functional Graph Distribution

CSES Problem Set**Empty String**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a string consisting of n characters between a and z.

On each turn, you may remove any two adjacent characters that are equal. Your goal is to construct an empty string by removing all the characters.

In how many ways can you do this?

Input

The only input line has a string of length n .

Output

Print one integer: the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 500$

Example

Input:

aabccb

Output:

3

CSES Problem Set

Permutation Inversions

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to count the number of permutations of $1, 2, \dots, n$ that have exactly k inversions (i.e., pairs of elements in the wrong order).

For example, when $n = 4$ and $k = 3$, there are 6 such permutations:

- $[1, 4, 3, 2]$
- $[2, 3, 4, 1]$
- $[2, 4, 1, 3]$
- $[3, 1, 4, 2]$
- $[3, 2, 1, 4]$
- $[4, 1, 2, 3]$

Input

The only input line has two integers n and k .

Output

Print the answer modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 500$
- $0 \leq k \leq \frac{n(n-1)}{2}$

Example

Input:

4 3

Output:

6

CSES Problem Set

Counting Bishops

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to count the number of ways k bishops can be placed on an $n \times n$ chessboard so that no two bishops attack each other.

Two bishops attack each other if they are on the same diagonal.

Input

The only input line has two integers n and k : the board size and the number of bishops.

Output

Print one integer: the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 500$
- $1 \leq k \leq n^2$

Example

Input:

5 4

Output:

2728

CSES Problem Set**Counting Sequences**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to count the number of sequences of length n where each element is an integer between $1 \dots k$ and each integer between $1 \dots k$ appears at least once in the sequence.

For example, when $n = 6$ and $k = 4$, some valid sequences are $[1, 3, 1, 4, 3, 2]$ and $[2, 2, 1, 3, 4, 2]$.

Input

The only input line has two integers n and k .

Output

Print one integer: the number of sequences modulo $10^9 + 7$.

Constraints

- $1 \leq k \leq n \leq 10^6$

Example

Input:

6 4

Output:

1560

CSES Problem Set

Grid Paths II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Consider an $n \times n$ grid whose top-left square is $(1, 1)$ and bottom-right square is (n, n) .

Your task is to move from the top-left square to the bottom-right square. On each step you may move one square right or down. In addition, there are m traps in the grid. You cannot move to a square with a trap.

What is the total number of possible paths?

Input

The first input line contains two integers n and m : the size of the grid and the number of traps.

After this, there are m lines describing the traps. Each such line contains two integers y and x : the location of a trap.

You can assume that there are no traps in the top-left and bottom-right square.

Output

Print the number of paths modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^6$
- $1 \leq m \leq 1000$
- $1 \leq y, x \leq n$

Example

Input:

3 1
2 2

Output:

2

CSES Problem Set

Counting Permutations

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A permutation of integers $1, 2, \dots, n$ is called *beautiful* if there are no adjacent elements whose difference is 1.

Given n , your task is to count the number of beautiful permutations.

Input

The only input line contains an integer n .

Output

Print the number of beautiful permutations of $1, 2, \dots, n$ modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 1000$

Example

Input:

5

Output:

14

CSES Problem Set

Grid Completion

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to create an $n \times n$ grid whose each row and column has exactly one A and B. Some of the characters have already been placed. In how many ways can you complete the grid?

Input

The first input line has an integer n : the size of the grid.

After this, there are n lines that describe the grid. Each line has n characters: . means an empty square, and A and B show the characters already placed.

You can assume that every row and column has at most one A and B.

Output

Print one integer: the number of ways modulo $10^9 + 7$.

Constraints

- $2 \leq n \leq 500$

Example

Input:

```
5
.....
..AB.
.....
B....
...A.
```

Output:

```
16
```

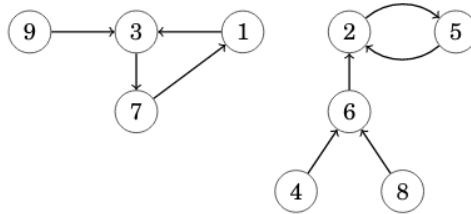
CSES Problem Set

Functional Graph Distribution

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A *functional graph* is a directed graph where each node has outdegree 1. For example, here is a functional graph that has 9 nodes and 2 components:



Given n , your task is to calculate for each $k = 1 \dots n$ the number of functional graphs that have n nodes and k components.

Input

The only input line has an integer n : the number of nodes.

Output

Print n lines: for each $k = 1 \dots n$ the number of graphs modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 5000$

Example

Input:

3

Output:

17

9

1